

Collaborators: _____

ws08 Alpha-beta pruning**Class Meeting 9 • 1 Oct 2025****Background**

Evaluating game tree nodes can be expensive. Alpha-beta pruning finds the correct minimax value without evaluating every node.

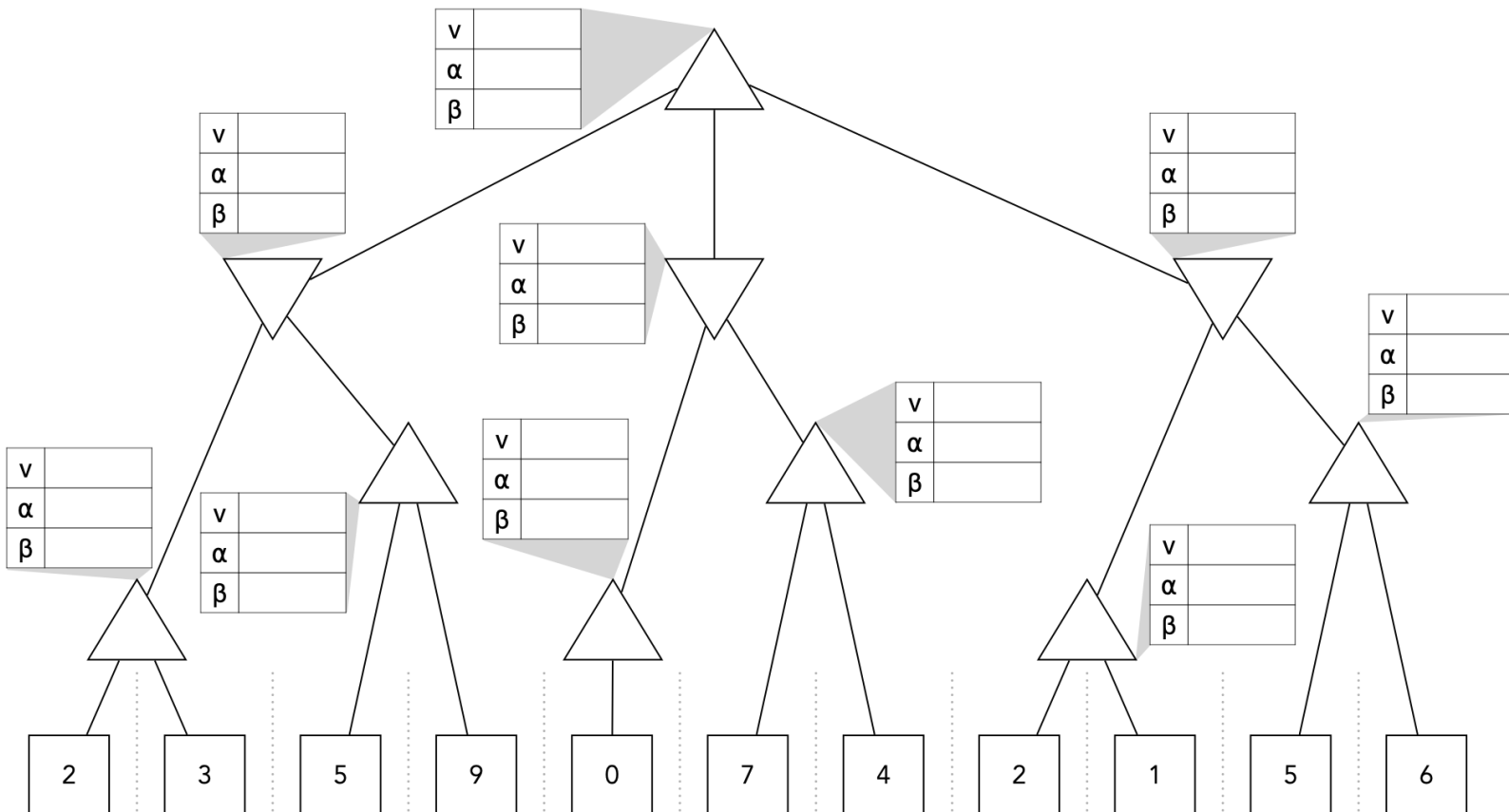
Consider the game tree shown below. $\triangle$ are maximizing nodes and ∇ are minimizing nodes. In this small tree, we will use alpha-beta pruning to find the minimax value of the top node *without* accessing every terminal utility ($\square$ nodes).

Tasks

- First, fold up the bottom of the sheet to cover the terminal nodes ($\square$). Tear off parts of the flap

to reveal terminal utilities *only* when queried by the algorithm.

- Execute the alpha-beta pruning algorithm, following the pseudocode provided. Start by calling `MAX-VALUE` on the top node with $\alpha = -\infty$ and $\beta = \infty$.
- Fill in and update the values of v , α and β for each node as you go.
- Mark pruned nodes or subtrees with an X. (You should end up pruning three times.)
- At the end, reveal all the terminal nodes and confirm that you found the correct minimax value.

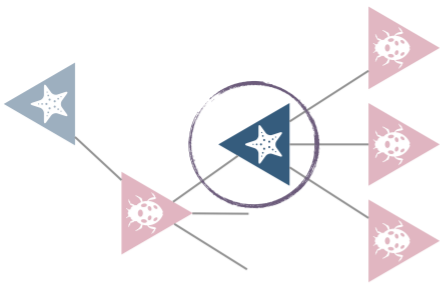


Fold up at the dashed line to conceal the terminal utilities ($\square$).

Alpha-beta pruning (MAX node)

1	function MAX-VALUE(state, α , β) returns an action	Finding the value of a MAX node
2	if TERMINAL-TEST(state) then return UTILITY(state)	The value of a terminal state is its utility
3	$V \leftarrow -\infty$	
4	for each a in ACTIONS(state) do	Iterate over successors
5	$s \leftarrow \text{RESULT}(\text{state}, a)$	
6	$v \leftarrow \text{MAX}(v, \text{MIN-VALUE}(s, \alpha, \beta))$	v will take on the maximum of the successors' values. (The successors are all MIN nodes)
7	if $v \geq \beta$ return v	If v is better than β , we don't need to keep going
8	$\alpha \leftarrow \max(\alpha, v)$	Is v the best option for MAX that we have seen?
9	return v	

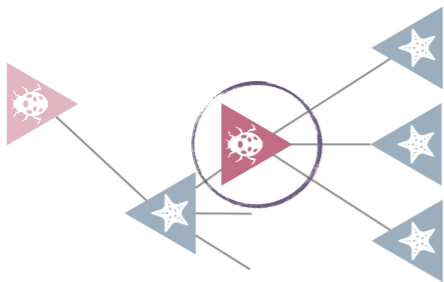
β the lowest value this node's MIN ancestors have access to



Why? Some MIN node above our MAX node would take the path to β , rather than come to this MAX node.

Alpha-beta pruning (MIN node)

1	function MIN-VALUE(state, α , β) returns an action	Finding the value of a MIN node
2	if TERMINAL-TEST(state) then return UTILITY(state)	The value of a terminal state is its utility
3	$v \leftarrow \infty$	
4	for each a in ACTIONS(state) do	Iterate over successors
5	$s \leftarrow \text{RESULT}(\text{state}, a)$	
6	$v \leftarrow \text{MIN}(v, \text{MAX-VALUE}(s, \alpha, \beta))$	v will take on the minimum of the successors' values. (The successors are all MAX nodes)
7	if $v \leq \alpha$ return v	If v is lower than α , we don't need to keep going
8	$\beta \leftarrow \min(\beta, v)$	Is v the best (lowest) option for MIN that we have seen?
9	return v	



Why? Some MAX node above our MIN node would take the path to α , rather than come to this MIN node.

α the best value this node's MAX ancestors have access to